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Roll No

ME-5004-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Dynamics of Machines

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume suitable data or dimensions, if necessary.
1. An IC Engine working on four stroke Otto cycle has a bore of 25 cm and stroke of 50cm. The length of connecting rod is 100cm and mass of reciprocating parts is 80 kg. The crank rotates at 240 rpm. The compression ratio is 5, the gas pressure at the end of constant volume heat addition is $30 \times 10^5 \text{ N/m}^2$, Suction pressure is $-0.1 \times 10^5 \text{ N/m}^2$, and exhaust pressure is $1.5 \times 10^4 \text{ N/m}^2$. The law of compression may be assumed as $PV^{1.34} = \text{constant}$, the Atmospheric pressure being $1.03 \times 10^5 \text{ N/m}^2$. Find the turning moment on Crank shaft when crank has turned 45° from IDC during power stroke.
 2. The Torque exerted on the crank shaft of a 2 stroke engine is given by the equation $T(\text{Nm}) = 7000 + 1000 \sin 2\theta - 2000 \cos 2\theta$ Where θ = crank displacement from IDC. Assuming the resisting torque to be constant. Determine:
 - a) The power developed when the Engine Speed is 300rpm
 - b) The total Fluctuations in speed in percentage
 - c) The Maximum Retardation of the Flywheel.The mass of Flywheel is 500 kg and its radius of Gyration is 750mm.

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3.
 - a) Define and explain the balancing of rotation masses. What will be the harm if the rotating parts of a high speed engine are not properly balanced?
 - b) Explain the method of direct and reverse cranks to determine the unbalance forces in radial engines.
4. The arms of a porter governor are 250 mm long. The upper arms are pivoted on the axis of rotation and the lower arms are attached to the sleeve at the sleeve at a distance of 40 mm from the axis of rotation. The load on the sleeve is 525 N and weight of each ball 75 N. Determine the equilibrium speed when the radius of the balls is 200 mm. What will be the range of speed for this position of the frictional resistance to the motion of sleeve is equivalent to 30W.
5.
 - a) Explain the balancing of two rotating masses in different planes.
 - b) Explain the reasons of unbalanced forces and couples in reciprocating machines.
6.
 - a) Which of the two assumptions uniform intensity of pressure or uniform rate of wear, would you make use of in designing friction clutch and why?
 - b) What is friction? Is it a blessing or curse? Justify your answer giving examples?
7.
 - a) Explain working of multi plate clutch.
 - b) A conical Pivot bearing supports a vertical shaft of 200mm diameter. It is subjected to a load of 30kN. The angle of the cone is 120° and co-efficient of friction is 0.025. Find the power lost in friction when the speed is 140rpm. Assuming.
 - i) Uniform pressure
 - ii) Uniform wear
8. Write short notes (any two)
 - a) Lanchester technique of Engine Balancing
 - b) Inertia Governor
 - c) Dynamometer

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